

José Miguel Ramírez Gutiérrez

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EDUCATION

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM) Querétaro, Mexico
B.S. in Robotics and Digital Systems Engineering (GPA: 3.7/4.0) Aug. 2024 – Expected 2028

- Socioeconomic Scholarship

EXPERIENCE

Cloud Integration & IT Specialist 2026 – Present
binarysails Querétaro, Mexico

- Collaborated to verify and standardize Microsoft OneDrive functionality, remote sharing, and synchronization across multiple external computers.
- Professionalized the delivery of project documentation by standardizing the organization of electronic and firmware files within shared cloud drives.

PROJECTS

Custom Linux Single Board Computer (SBC) | KiCad, Allwinner V3s, Hardware Design July 2026

- Engineered a custom 4-layer PCB around the Allwinner V3s (ARM Cortex-A7) processor, integrating an EA3036CQB PMIC, Ethernet PHY, and onboard RTC.
- Implemented differential pair routing with controlled impedance to guarantee signal integrity for High-Speed USB 2.0 (480 Mbps) communication.
- Optimized board layout and component placement for automated SMT manufacturing (PCBA), resolving centroid rotation discrepancies and footprint constraints.

Automated Key Fob Positioning System (PEPS Validation) | Python, Arduino, C++ June 2026

- Designed and implemented CoreXY kinematics firmware in C++ with non-blocking stepper control, trapezoidal acceleration profiles, and a serial command protocol for remote operation.
- Built an autonomous workspace calibration system using 4 HC-SR04 ultrasonic sensors with a halving-approach algorithm, achieving ± 1 cm positioning accuracy over a 390×1300 mm area.
- Developed a Python/Tkinter GUI for real-time control, programmable movement sequences with EEPROM persistence, configurable origin selection, and emergency stop handling.

ESP32 AI Voice Assistant | MicroPython, KiCad May 2026

- Designed a custom PCB in KiCad integrating an ESP32, INMP441 I2S microphone, PAM8403 amplifier, and SH1106 OLED display; generated and ordered production-ready Gerber files.
- Achieved rapid response times by integrating Groq API (Whisper + LLaMA 3.1) for audio transcription and LLM processing via a Flask backend deployed on Render.
- Implemented wake-word detection with VAD-based audio capture.

DetectaGas – IoT Gas Leak Prevention System | ESP32, Rust, React, Axum December 2025

- Engineered an IoT device using an ESP32 and MQ-5 sensor to rapidly reduce response time to hazardous LP gas leaks.
- Built a concurrent Rust backend with Axum and a React web interface, processing continuous telemetry via an MQTT broker.

Flight Monitoring & Data Analysis | Python January 2026

- Processed and cleaned structured datasets of commercial flights using Python, improving overall trend visualization accuracy.
- Developed modular scripts for scalability, drastically reducing data preprocessing time for real-world analytical tasks.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, C#, JavaScript, MATLAB, VHDL

Embedded & Hardware: ESP32, Arduino, KiCad (PCB Design)

Software & Technologies: MicroPython, Arduino IDE, Axum, MQTT, MySQL, Git, React, Flask

Spoken Languages: Spanish (Native), English (Business Fluent)